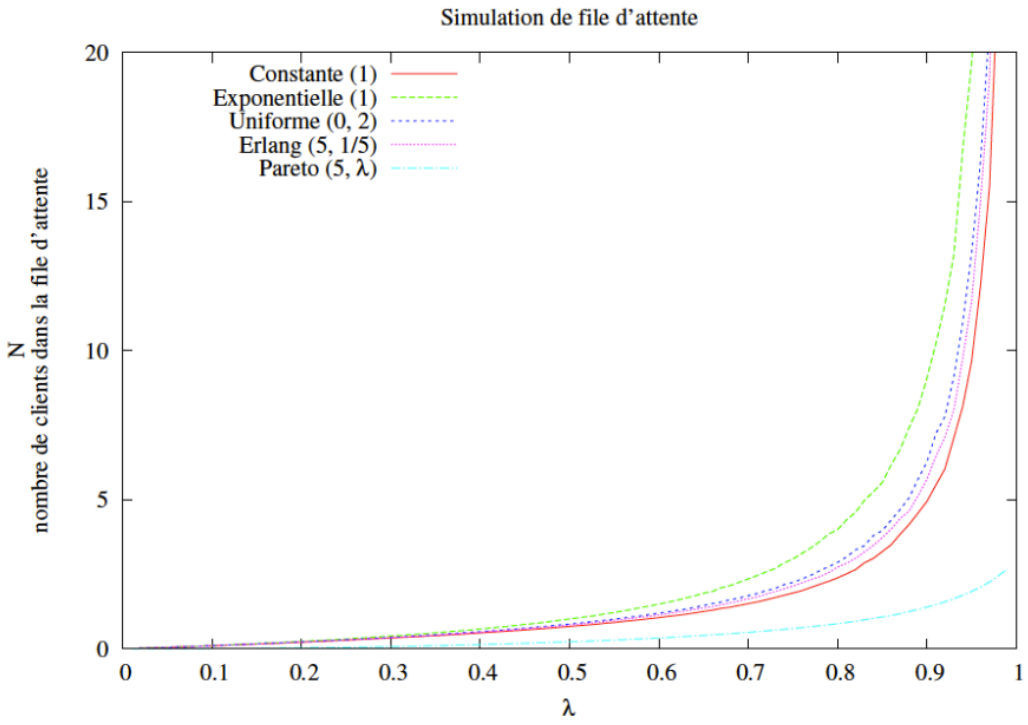


FIGURE 1 – Impact de la variabilité du temps de service sur le nombre moyen de clients dans la file d'attente

Figure: slide 45

Good

- A line plot is good choice
- Both axes are labelled, the origin is (0,0)
- The legend is in the top-left corner, it does not hide any curve and it is easy to see
- There are five curves all share the same y-axis

Problems

- The caption says that the figure shows the mean number of customers, but the y-axis does not say "mean" (Annotations)
- The Constante, Uniforme and Erlang curves are almost on top of each other, so it is hard to tell them apart. (Graphical objects)
- There is no grid, so it is hard to read values. (Graphical objects)
- There are no confidence intervals, and we do not know how many times the simulation was repeated. (Data, Information)

How to improve

- Write “Mean number of customers in the queue” on the y-axis, so the reader knows that N is an average.
- Change the range of the x-axis or the y-axis, so that all the curves can be seen well
- Add a light grid.
- Repeat the simulation several times and draw 95% confidence bands. Write the number of repetitions in the caption.

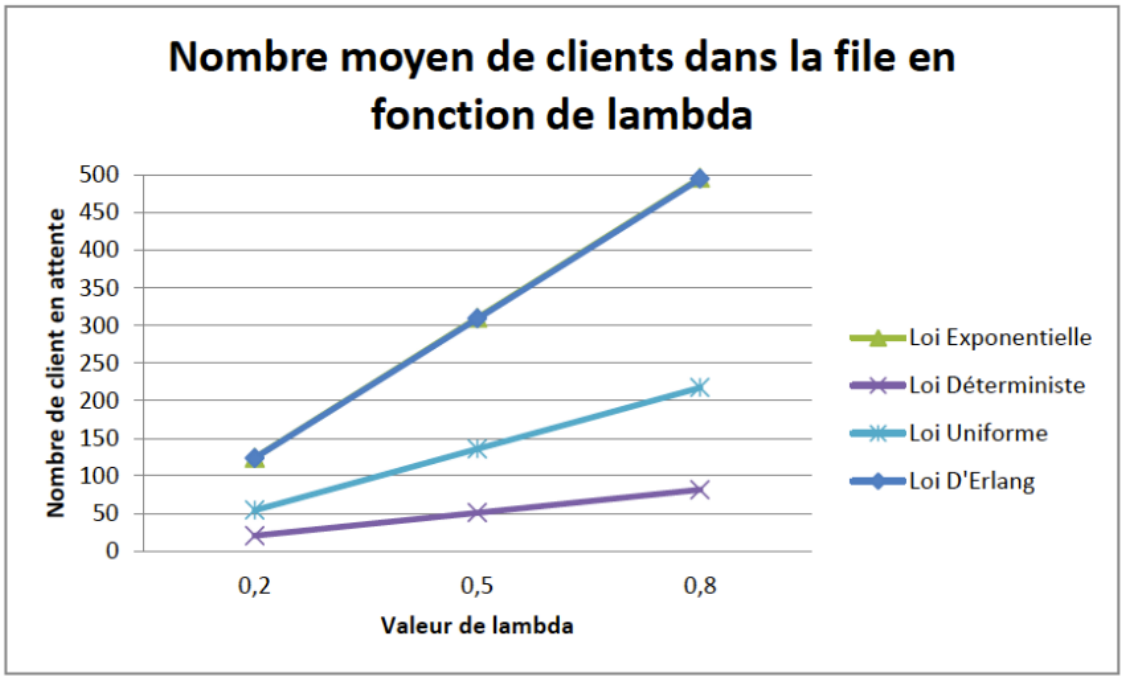


Figure: slide 40

Good

- A line plot is a good choice
- The figure has a title, both axes are labelled, and the y-axis starts at 0
- There are four curves, each with a distinct marker

Problems

- Two of the lines are exactly on top of each other, so it is very hard for the reader to find them. (Graphical objects)
- Each curve has only three points, which is not enough to show the real trend. (Data)
- There are no error bars or confidence intervals, and we do not know how many times the simulation was repeated. (Data, Information)

How to improve

- Use more lambda values on a numeric axis.
- Use distinct colour-blind-safe colours and line types, and label curves directly at their ends.
- Repeat the simulation several times and show the error range (95% confidence interval) on the graph.

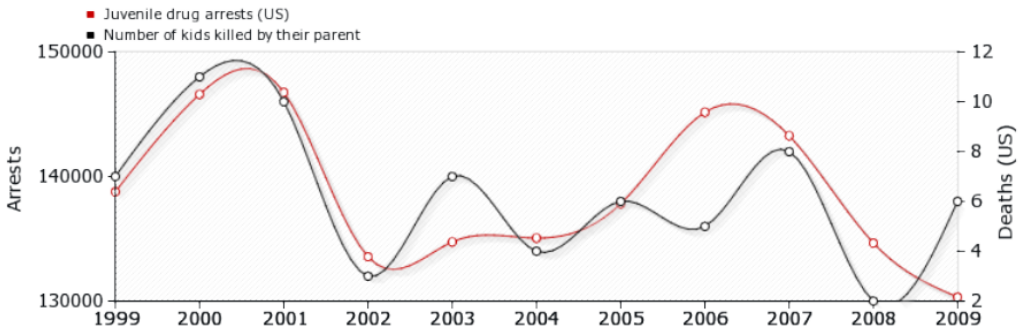


Figure: slide 33

Good

- The data points are marked with circles, so we can see the yearly values.
- There are only two curves, and the figure is not overloaded.
- The two curves have different colours

Problems

- There are two y-axes with different scales (130,000 to 150,000 and 2 to 12). This makes two unrelated variables look almost the same. (Information)
- The smooth curves go through only 11 points. This kind of interpolation does not make sense, and around 2008 the curve even goes below the axis. (Data)
- Neither y-axis starts at 0, and there is no explanation for it. (Annotations)
- There is no title, no data source, and the variables are not defined. (Context)

How to improve

- Use two small plots separately.

- Add a title that makes the figure easy to understand, the data source, and explain what each variable is.
- Join the points with straight lines instead of smoothing.